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Autosomal Dominant Cerebellar Ataxia, Deafness, and Narcolepsy (ADCA-DN) Associated With Progressive Cognitive and Behavioral Deterioration

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Objective: Autosomal dominant cerebellar ataxia with deafness and narcolepsy (ADCA-DN) is an emerging syndrome caused by mutations in the C-terminus end of the TS domain of the *DNMT1* gene. ADCA-DN is also associated with sensorimotor polyneuropathy, extrapyramidal, and dysautonomic signs, as well as dementia. Little has been reported about the progressive cognitive impairment associated with ADCA-DN. Our objective is to provide a detailed characterization of the cognitive profile of ADCA-DN. **Method:** Three members of a kindred with ADCA-DN underwent comprehensive neuropsychological testing and neuroimaging. **Results:** At baseline, 2 individuals demonstrated cognitive profiles with executive difficulties in some areas consistent with frontal-system dysfunction behaviorally and on standardized testing. The third individual was further in the disease course and exhibited more globally impaired cognition consistent with a diagnosis of dementia. **Conclusions:** This family demonstrated progressive neurodegeneration beginning with isolated areas of executive dysfunction and leading to globally impaired cognition and dementia. Cognitive decline occurred in parallel with neurological deterioration. The cognitive profile is similar to case reports of other individuals with an allelic neurological phenotype, Hereditary Sensory Autonomic Neuropathy 1E, also caused by *DNMT1* mutations.

Keywords: ADCA-DN, HSAN 1E, DNMT1, neuropsychology, cognition

Autosomal dominant cerebellar ataxia with deafness and narcolepsy (ADCA-DN) is an increasingly recognized adult-onset progressive neurodegenerative disorder first described in 1995 in a Swedish

pedigree comprising five affected family members (Melberg et al., 1995). The condition has since been documented in several other kindreds (Moghadam, Pizza, Morgia et al., 2014; Moghadam, Pizza, Tonon et al., 2014; Pedroso et al., 2013), not including the current family. ADCA-DN is an autosomal dominant inheritance disease, with cases of sporadic mutations (Pedroso et al., 2013). Symptoms of ADCA-DN include: narcolepsy-cataplexy beginning typically in the 30s and 40s, sensory neuronal deafness, cerebellar ataxia, and dementia. More variable symptoms such as psychosis, optic atrophy, and diabetes mellitus can also be present (Melberg et al., 1995; Winkelmann et al., 2012). Exome sequencing revealed that exon 21 mutations in *DNMT1* cause ADCA-DN (Winkelmann et al., 2012). DNMT1, or DNA methyltransferase 1, is a key methylation enzyme that is required during DNA replication and DNA repair (Klein et al., 2013). The DNA methylation process alters gene expression during cell differentiation into specific tissues. DNMT1 is particularly important for the maintenance of cytosine methylation at the cytosine-phosphate-guanine dinucleotides in the mammalian genome (Feng & Fan, 2009). DNMT1 is vital for gene regulation and chromatin stability and is critical for the development, survival and connectivity of neural cells (Chen et al., 2003; Tohgi et al., 1999).

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We would like to extend our sincere thanks to the three family members of the ADCA-DN kindred discussed here. Their time and effort in the completion of these investigations are much appreciated. We are in awe of their courage and determination.

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DNMT1 mutations also cause a similar adult onset disorder: Hereditary Sensory Autonomic Neuropathy with dementia and hearing loss (HSAN1E). Genetic mutations causal for ADCA-DN are located in the C-terminus end of the TS domain, whereas mutations for HSAN1E are located in the middle part or N-terminus end of the TS domain of the *DNMT1* gene (Baets et al., 2015). Sensory neuropathy and deafness are typically the first symptoms with dementia and cerebellar ataxia present as the condition progresses (Klein et al., 2011). In affected individuals, childhood development proceeds normally and symptoms of hearing loss and sensory neuropathy do not become apparent until the teens or mid-30s. The initial clinical manifestations are followed by memory decline in the 30s and 40s. Affected individuals typically require total care and institutionalization by the 5th decade as a result of the memory loss, sensory ataxia, and foot ulcers or extremity amputations (Klein et al., 2013, 2011; Moghaddam, Pizza, Morgia et al., 2014). Autopsy studies of individuals with HSAN1E documented diffuse neurodegeneration of the brain, spinal cord, and nerve fibers but no interstitial changes (i.e., no β -amyloid, TDP-43, or α -synuclein immunostaining; Klein et al., 2013).

Limited neurocognitive testing of the patients with *DNMT1* conditions have been reported, but these have been primarily focused on patients with overlapping HSAN1E phenotype. Wright and Dyck (1995) administered a limited neuropsychological battery to a 42-year-old man with HSAN1E. Evaluation with the WAIS-R, WMS-R and Boston Naming Test revealed impairment in general intellectual abilities, attention, constructional praxis, language, learning, memory, and abstract problem solving. The proband's elder brother exhibited a similar clinical picture with more severe dementia and loss of independence in the 5th decade. An additional case was described involving a male with cognitive impairment and "odd behaviors" beginning in his 50s with diffuse cerebral and cerebellar atrophy on magnetic resonance imaging (MRI) of the brain (Klein et al., 2013). Cognitive testing was limited to the MMSE (23/30) with a description that memory was better than executive functioning and judgment. He was given a diagnosis of probable frontotemporal dementia. The proband's 61-year-old brother also demonstrated mild to moderate cognitive impairment in the context of mild diffuse cerebral atrophy on MRI brain (Klein et al., 2013). An additional kindred included a 47-year-old woman with sensory neuropathy, foot ulcers, mild deafness, and cognitive problems. She obtained a Montreal Cognitive Assessment (MoCA) score of 24/30 with "forgetfulness" and was diagnosed with hoarding and obsessive-compulsive disorder. An additional cohort of patients with HSAN1E phenotype and cognitive impairment is also described that includes an adult male and female siblings in the fourth decade with isolated deficits in verbal, attention, or logical tasks, as well as a 59-year-old male with a 25 year history of symptoms with global cognitive impairment with verbal and visuospatial dysfunctions (Moghaddam, Pizza, Morgia et al., 2014).

These limited, cognitive assessments have been primarily restricted to the HSAN1E phenotype. However, no group has yet comprehensively characterized and published a cognitive profile of the ADCA-DN clinical phenotype. The Ottawa Neurogenetics Clinic identified a family with ADCA-DN. Several family members have undergone detailed genetic analysis and six individuals

were identified as likely affected based on examination and chart review. The four living symptomatic family members were identified to have the pathogenic *DNMT1* mutation NM_001130823.1:c.1709C>T (exon21); NP_001124295.1:p.Ala570Val. The children of the current probands are underage and have not undergone genetic testing and to date remain unaffected (see Figure 1). Of the four identified affected living family members, three agreed to undergo comprehensive neuropsychological testing. The current objective was to characterize the cognitive profile in each of these three individuals with ADCA-DN because of *DNMT1* mutations.

Case Reports

Results of genetic testing revealed that all three patients were confirmed to have the *DNMT1* mutation associated with ADCA-DN, NM_001130823.1:c.1709C>T (exon21); NP_001124295.1:p.Ala570Val, which is a previously reported pathogenic mutation (Winkelmann et al., 2012).

Patient 1

Patient 1 was a 41-year-old married gentleman with three sons. He has a Grade 12 education and was previously employed as a drywall installer/painter but was receiving long-term disability benefits because of progressive ataxia and hearing loss at time of assessment. Past medical history includes juvenile myoclonic epilepsy, diagnosed as a teen-ager, well-controlled with valproic acid with no seizures for over a decade. At age 25, he complained of headaches and decreased memory, and had a "normal" neurological examination and computed tomography (CT) head. At age 40, he was assessed as part of the familial cohort seeking genetic testing for the *DNMT1* gene, with a 5 year history of progressive bilateral sensorineural hearing loss. Additional symptoms included mild gait ataxia (presenting as minimal difficulty in tandem gait), excessive daytime sleepiness, intermittent tremors of the right hand/arm, hyperhidrosis, mild cognitive changes, and it was noted that the patient exhibited a "lively/slightly disinhibited demeanor." Nerve conduction studies at age 40 were normal. MRI of the brain documented prominent cortical sulci in the frontal and parietal lobes over the high convexities, and possible mild volume loss. Patient 1 reported that he drinks 10 alcoholic drinks per week (more on social occasions). Alcohol consumption had been greater 10 years ago (30 to 40 drinks per week). Although the contribution of his excessive alcohol consumption cannot be fully ruled out as a contributing factor in his neuropsychological profile, his young age, the reduction in his con-

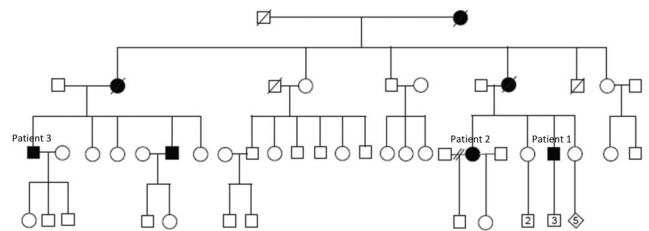


Figure 1. Family pedigree demonstrating autosomal dominant inheritance.

Table 1
Baseline Neuropsychological Assessment Results

Test	Patient 1 baseline ^a	Patient 2 baseline ^a	Patient 3 baseline ^a
Intellectual functioning (WASI)			
Verbal IQ	27	27	47
Performance IQ	67	68	n/a
Full Scale IQ	50	50	n/a
Similarities	27	62	69
Vocabulary	31	8	27
Block Design	66	58	7
Matrix Reasoning	73	79	n/a
Attention and information processing speed			
Digit Span forward	62	13	17
Digit Span backward	27	56	36
Mental Control	50	63	16
Symbol Digit Modalities Test	38	66	n/a
Auditory Consonant Trigrams	9 sec 21	5	n/a
Auditory Consonant Trigrams	18 sec 21	14	n/a
D-KEFS Trails 1–Visual Scanning	63	50	<1
D-KEFS Trails 2–Number Sequencing	75	75	<1
D-KEFS Trails 3–Letter Sequencing	63	63	<1
D-KEFS Trails 5–Motor Speed	75	24	16
Sensory motor abilities			
Grooved Pegboard	Dominant hand Nondominant hand	<1 16	d/c (imp) n/a
Language			
Boston Naming Test	3	1	<1
D-KEFS Verbal Fluency	Letter Category	37 84	1 2
Visual perceptual abilities			
Clock drawing	WNL	WNL	imp
House drawing	WNL	Mildly ↓	Mildly ↓
Cube drawing	WNL	Mildly ↓	WNL
Hooper Visual Organization Test	WNL	WNL	n/a
Rey Complex Figure Test	Copy Time to Copy	>16 >16	n/a n/a
Judgement of Line Orientation	58	38	n/a
Memory			
WMS-IV	Logical Memory immediate Logical Memory delayed	91 91	n/a n/a
CVLT-II	Total List B	38 50	1 16
	Short Delay Free	50	7
	Short Delay Cued	16	2
	Long Delay Free	50	7
	Long Delay Cued	50	2
	Recognition	31	70
BVMT-R	Total	69	14
	Delay	90	38
RCFT	Immediate	18	n/a
	Delay	21	n/a
	Recognition	76	n/a
Executive abilities			
Wisconsin Card Sorting Test	Categories Perseverative responses Perseverative errors	>16 21 21	n/a n/a n/a
	Loss of set	2–5	n/a
D-KEFS Verbal Fluency	Switching Total Correct Switching Accuracy	9 16	1 1
D-KEFS Color-Word	Color Word	63 37	<1 1
	Inhibition	50	<1
	Switching	50	<1
D-KEFS 20 Questions	Abstraction	50	n/a
	Total Questions	75	n/a
	Achievement	84	n/a

(table continues)

Table 1 (continued)

Test		Patient 1 baseline ^a	Patient 2 baseline ^a	Patient 3 baseline ^a
D-KEFS Word Context	Total Consecutive Correct	84	63	5
D-KEFS Tower	Total Achievement	75	n/a	n/a
	Time-per-move	63	n/a	n/a
D-KEFS Trails	Number-Letter Switching	5	75	<1
Mood				
Beck Depression Inventory–Fast Screen		Minimal	Mild	Minimal
State-Trait Anxiety Inventory	State	5	81	21
	Trait	21	65	6

Note. WASI–Wechsler Abbreviated Scale of Intelligence; D-KEFS–Delis-Kaplan Executive Function System; WNL–within normal limits; WMS-IV–Wechsler Memory Scale–IV; CVLT-II–California Verbal Learning Test–II; BVMT-R–Brief Visuospatial Memory Test–Revised; RCFT–Rey Complex Figure Test; n/a–not administered; d/c–discontinued; imp–impaired.

^a Scores in percentiles except where indicated.

sumption, and the fact that his deficits (see below) were no worse than Patient 2, argue that any impact of alcohol was likely negligible.

Patient 1 did not report any cognitive complaints at the time of the baseline neuropsychological evaluation. In contrast, his wife commented that she had noticed diminished memory (although not to the point of concern), increased irritability and impatience. She also noted that he was less outgoing in social situations, which she attributed to his hearing loss.

During the follow-up evaluation, Patient 1 (age 42) noted further hearing decline to the point that he was almost completely deaf without his hearing aids. With the hearing aids and with his ability to supplement his hearing with lip reading, he was able to engage in appropriate conversation and could follow all test instructions without difficulty. As such, there was no concern that his hearing impairment impacted his performance. He also demonstrated an increase in bilateral postural arm tremors. He reported no progression in behavior, cognition, sleep, or balance. In contrast, his wife noted that he was easily angered, with reduced self-awareness of his behavioral problems, and as a result had strained relations with some friends and family.

Patient 2

Patient 2 was a 47-year-old married woman with one daughter and one son. She had a Grade 12 education, was receiving disability benefits because of cognitive deficits and narcolepsy and was previously employed as a cashier. Medical history included hypothyroidism, pregnancy-induced hypertension previous placental failure during her first pregnancy. Baseline medications included, levothyroxine, esomeprazole, and modafinil.

At age 39, she presented with bilateral postural hand tremor and deteriorating handwriting quality. Neurological exam demonstrated sensorineural hearing loss, diminished balance, mild postural tremor and tandem gait ataxia, minimal nystagmus at extremes of horizontal gaze, and mild dysmetria. Polysomnographic studies at age 42 were consistent with narcolepsy. Over repeat examinations, she was documented to have a worsening clinical exam demonstrating progressive deafness bilaterally, postural tremor with worsening sleepiness/sleep attacks, sleep paralysis (awakening with transient global quadriplegia), gait ataxia, and nystagmus. She was described as having a “very happy/enthusiastic disposition.” Cognitive difficulties were re-

ported by her husband including poor short-term memory (STM), diminished attention, and problem solving. Nerve conduction studies at age 46 were consistent with length-dependent axonal polyneuropathy, although the patient was asymptomatic. MRI revealed global cerebral and cerebellar volume loss.

On neuropsychological testing, at age 47, the patient reported mild cognitive difficulties with complicated multistep tasks and decreased short term memory. Her husband described more significant cognitive and behavioral dysfunction including perseveration, poor source memory, tangential thoughts, incorrect recall of events, decreased frustration tolerance, and emotional lability.

Patient 3

Patient 3 was a 57-year-old with a teacher’s college education and additional training as a financial analyst. He was employed as a carpenter/carpentry teacher and then financial analyst, and at the time of the assessment was on long-term disability because of progressive cognitive impairment. He was married with three children. Medical history included gallstone pancreatitis, dyslipidemia, hypertension, and squamous cell carcinoma of the tongue treated with surgical resection and radiation. Chart review revealed severe bilateral sensory neuronal hearing loss for 25 years, narcolepsy/cataplexy, poor balance for 5 years, moderate gait ataxia, and loss of vibration and pin/pain perception in both feet. Cognitive difficulties were reported as STM loss and difficulty with new/complex tasks. At age 56 he scored 27/30 on the MoCA with difficulties noted on the cube, word fluency and abstraction items. MRI revealed generalized cerebral volume loss. Nerve conduction studies revealed moderate to severe length-dependent axonal sensory polyneuropathy.

At baseline neuropsychological evaluation at age 57 Patient 3 could not articulate any specific cognitive difficulties. However, his wife described slower cognitive processing, word finding difficulties, and decreased ability to perform multistep tasks. She also described decreased attention with distraction and emotional lability, which were denied by the patient. Despite his complex medical history, we do not believe the additional conditions were significantly contributory with regard to his cognitive profile. His hypertension was controlled with medication and although an MRI of the brain revealed generalized volume loss, there was no mention of microvascular disease. Similarly, the radiation he received to

Table 2

Baseline and Follow-Up Neuropsychological Assessment Results for Patient 1

Test		Patient 1 baseline ^a	Patient 1 follow-up ^a
Intellectual Functioning (WASI)			
Verbal IQ		27	25
Performance IQ		67	70
Full Scale IQ		50	50
Similarities		27	21
Vocabulary		31	34
Block Design		66	68
Matrix Reasoning		73	73
Attention and Information Processing Speed			
Digit Span forward		62	32
Digit Span backward		27	50
Mental Control		50	25
Symbol Digit Modalities Test		38	53
Auditory Consonant Trigrams	9 sec	21	2
Auditory Consonant Trigrams	18 sec	21	18
D-KEFS Trails 1–Visual Scanning		63	50
D-KEFS Trails 2–Number Sequencing		75	75
D-KEFS Trails 3–Letter Sequencing		63	37
D-KEFS Trails 5–Motor Speed		75	50
Sensory Motor Abilities			
Grooved Pegboard	Dominant hand	1	14
	Nondominant hand	1	1
Language			
Boston Naming Test		3	12
D-KEFS Verbal Fluency	Letter	9	9
	Category	16	9
Visual Perceptual Abilities			
Clock drawing		WNL	WNL
House drawing		WNL	WNL
Cube drawing		WNL	WNL
Hooper Visual Organization Test		WNL	WNL
Rey Complex Figure Test	Copy	>16	>16
	Time to Copy	>16	>16
Judgement of Line Orientation		58	75
Memory			
WMS-IV	Logical Memory immediate	84	63
	Logical Memory delayed	84	63
CVLT-II	Total	69	53
	List B	16	68
	Short Delay Free	50	32
	Short Delay Cued	69	50
	Long Delay Free	50	16
	Long Delay Cued	50	16
	Recognition	50	50
BVMT-R	Total	62	3
	Delay	90	18
RCFT	Immediate	86	54
	Delay	88	62
	Recognition	31	90
Executive Abilities			
Wisconsin Card Sorting Test	Categories	2–5	6–10
	Perseverative responses	2	6
	Perseverative errors	1	8
	Loss of set	>16	6–10
D-KEFS Verbal Fluency	Switching Total Correct	37	37
	Switching Accuracy	16	50
D-KEFS Color-Word	Color	37	9
	Word	24	2
	Inhibition	50	63
	Switching	50	37
D-KEFS 20 Questions	Abstraction	84	37
	Total Questions	50	91
	Achievement	63	84

(table continues)

Table 2 (continued)

Test		Patient 1 baseline ^a	Patient 1 follow-up ^a
D-KEFS Word Context	Total Consecutive Correct	84	n/a
D-KEFS Tower	Total Achievement	75	84
	Time-per-move	63	75
D-KEFS Trails	Number-Letter Switching	5	37
Mood			
Beck Depression Inventory–Fast Screen		Minimal	Minimal
State-Trait Anxiety Inventory	State	5	37
	Trait	21	n/a

Note. WASI–Wechsler Abbreviated Scale of Intelligence; D-KEFS–Delis-Kaplan Executive Function System; WNL–within normal limits; WMS-IV–Wechsler Memory Scale-IV; CVLT-II–California Verbal Learning Test-II; BVM-T-R–Brief Visuospatial Memory Test–Revised; RCFT–Rey Complex Figure Test; n/a–not administered; d/c–discontinued; imp–impaired.

^a Scores in percentiles except where indicated.

treat his cancer was localized to the tongue and no generalized effect on the brain is expected.

Method

Patient 1 (male), his sister Patient 2 (female), and their cousin Patient 3 (male) underwent comprehensive neuropsychological evaluations (the scope of Patient 3's evaluation was limited by his visual impairment, level of fatigue, and severity of cognitive impairment). The tests were selected to establish functioning in all cognitive domains (see Tables 1 and 2). Patient 3 was unable to complete all tasks administered to Patients 1 and 2 given the severity of his condition. Self-report questionnaires evaluated symptoms of depression and anxiety. Finally, all three patients and their spouses completed the Frontal Systems Behavior Scale (FrSBe) to evaluate their perception of the presence of behaviors associated with frontal system dysfunction both before and after the onset of illness-related symptoms. The three subscales of the FrSBe (apathy, disinhibition, and dysexecutive behaviors) are designed to correspond to the three frontal behavioral syndromes that can occur as a result of damage to the three fronto-striato-thalamic circuits (Grace & Malloy, 2001). The apathy subscale corresponds to the anterior cingulate circuit and evaluates behaviors such as diminished motivation and activation, abulia, and diminished spontaneity. The disinhibition subscale corresponds to the orbital prefrontal circuit and reflects problems such as behavioral dysregulation, impulsivity, and poor self-monitoring. Finally, the executive subscale corresponds to the dorsolateral prefrontal circuit and taps into the more cognitive aspects of executive dysfunction such as diminished working memory, problems with multitasking, poor problem solving, and organizational difficulties.

Neuroimaging

Patient 1 and 2 underwent functional MRI (fMRI), performed on a 1.5 T Siemens Magnetom Symphony scanner. Patient 3 was not appropriate for fMRI given his inability to consistently remain alert for the initial cognitive testing. For Patients 1 and 2, a whole brain echo planar gradient echo pulse sequence (TR/TE 3,000/40 ms, flip angle 90 degrees, FOV 250 × 187.5 mm, 64 × 64 matrix, slice thickness 5 mm, 27 axial slices, bandwidth 2430 Hz per pixel) was used for the fMRI data acquisition. Two cognitive

tasks were performed for each participant: a Go/No Go task of response inhibition and a letter 2-back working memory task. White letters were presented on a black screen, one at a time and instructions were provided to guide the participants' responses. The Go/No Go task included a 'Press for X' baseline condition and a 'Press for all letters except X' test condition. The two-back task included a 'Press for X' baseline condition and a 'Press for 2-Back' test condition. For both tasks, the images acquired during the baseline condition were subtracted from those acquired during the test condition, leaving only the neural activity related to the type of processing of interest, namely, response inhibition and working memory.

DTI images were acquired to investigate white matter integrity. A spin-echo echoplanar parallel GRAPPA diffusion weighted imaging sequence was used with an acceleration factor of 2, 12 noncollinear directions and 2 b-values (0, 600 s/mm²) and 2 × 2 × 7 mm³ voxel resolution. Fractional anisotropy (FA) and mean diffusivity (MD) were calculated.

Follow-Up

Patient 1 returned for a follow-up neuropsychological evaluation after 19.5 months. Patient 2 was offered a follow-up evaluation but declined given ongoing psychosocial issues related to her progressive neurocognitive dysfunction. Patient 3 was not appropriate for a follow-up evaluation given his inability to consistently remain alert for the initial testing. Genetic testing, neurological examination, nerve conduction studies, and structural MRI brain imaging was done clinically with findings reported above.

Results

Behavioral Observations

All three individuals were friendly and rapport was easily established. Mild to moderate hearing loss was evident in all three individuals and instructions occasionally required repetition. Patients 1 and 2 demonstrated a mild tremor on writing/drawing tasks; whereas Patient 3 demonstrated more significant motor control difficulties in his hands (see Grooved Pegboard). Patients 1 and 2 were alert, with no obvious gait ataxia, and no gross perceptual deficits. In contrast, Patient 3 demonstrated obvious

narcoleptic episodes and drowsiness, marked gait ataxia, and marked perceptual deficits (e.g., he had to be guided to the door of the office). Patients 1 and 2 appeared mildly but inappropriately jovial during testing. Patient 3 was more reserved but engaged in occasional excessive laughter. None exhibited any gross disinhibition, though Patient 1 was mildly impulsive on testing. Effort was not formally assessed but no concerns were apparent on embedded measures and all were motivated to perform well.

Neuropsychological Test Results

See Table 1 for baseline cognitive scores on all three patients and to Table 2 for the baseline and follow-up scores on Patient 1. Table 3 lists mean *z*-scores by domain for all three patients at baseline.

Given the predominance of behavioral difficulties noted on the FrSBe, coupled with some of the findings on cognitive testing, executive dysfunction seems to be a prominent aspect of the presentation of these individuals. Behavioral dysfunction was exhibited on testing in the form of an inappropriately jovial manner, as well as impulsivity (in Patient 1) and perseveration (in Patients 1 and 3). Spouse responses to the FrSBe, a questionnaire designed to evaluate frontal system dysfunction, revealed that all three patients demonstrated behavioral difficulties in the form of apathy (i.e., diminished initiative, emotional unresponsiveness, and less engaged) and disinhibition (i.e., occasional inappropriate comments, embarrassing behavior, insensitivity to others' needs, and reduced frustration tolerance and short fuse). Patient questionnaires also revealed the more cognitive aspects of executive dysfunction (i.e., unawareness of difficulties, difficulty focusing in the context of distractions, difficulties planning ahead, inflexibility, and difficulties benefiting from feedback; Figures 2, 3, and 4). Of note is that in contrast to Patient 1's wife's verbal report of increased behavioral difficulties at the time of follow-up, both Patient 1's self and family ratings of behavioral difficulties were largely lower at follow-up compared to baseline. It is likely that this is because of the fact that at follow-up they were asked only to rate behavior "now" rather than to rate behavior before and after illness as they did at baseline. As such, they did not have the preillness comparator for reference.

On objective cognitive measures, performance on formal measures of executive functioning was more variable. Patient 1 was perseverative and impulsive and also demonstrated difficulties with concept formation and the ability to adapt in response to feedback. Difficulties with mental shifting were noted on both the Wisconsin Card Sorting Test (WCST) and

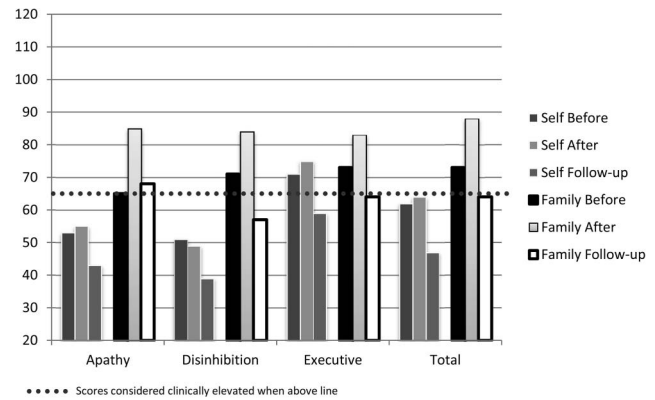


Figure 2. Frontal Systems Behavior Scale – *T*-scores before and after illness onset for Patient 1.

D-KEFS Trails (Number-Letter Sequencing). Although not impaired, working memory under conditions of interference was low average and thus fell below expected limits. Mild vulnerability to the effects of proactive interference was noted by his low average performance on List B on the California Verbal Learning Test–II. Lastly, verbal initiation was mildly diminished and fell in the borderline range. Other aspects of executive functioning fell within normal limits in Patient 1 including the inhibition of a primed response, deductive reasoning (on a 20 Questions task), verbal abstraction, and planning. Patient 2 demonstrated a tendency to lose mental set, mild perseveration (low average), borderline to low average mental shifting on a switching task, as well as borderline to low average working memory under conditions of interference. Areas of executive functioning that fell within normal limits in Patient 2 included concept formation, rapid mental shifting, inhibition of a primed response, abstraction, problem solving, and deductive reasoning. Patient 3's executive functioning was more severely impacted. He was too impaired to complete the Wisconsin Card Sorting Test or D-KEFS 20 Questions. Similarly, he was too impaired to complete a complex working memory task (under conditions of interference). On completed tasks he demonstrated impairment in initiation and maintenance of behavior, inhibition, rapid mental shifting, and conceptual abilities. When considering other aspects of cognition, each patient also demonstrated prominent word-finding difficulties and diminished fine motor abilities bilaterally. Patient 3 was more severely

Table 3
Mean *z*-Scores by Domain

Domain	Patient 1 baseline	Patient 2 baseline	Patient 3 baseline
Intellectual Functioning (WASI)	-.04	-.03	-.41
Attention and Information Processing Speed	-.01	-.26	-1.76
Sensory Motor Abilities	-2.30	-2.00	n/a
Language	-1.42	-.55	-2.45
Visual Perceptual Abilities	.07	-.10	n/a
Memory	.38	.14	-1.24
Executive Abilities	-.41	-.22	-2.66

Note. WASI–Wechsler Abbreviated Scale of Intelligence.

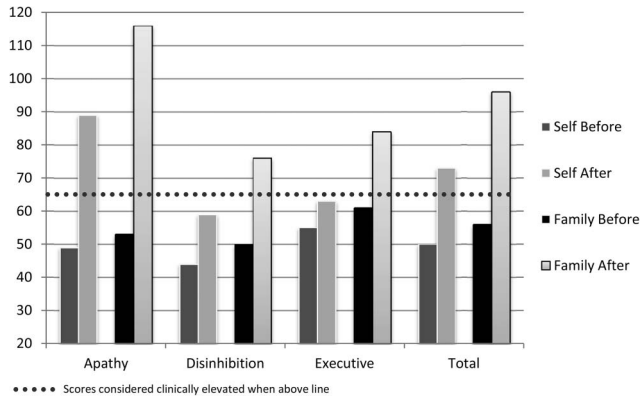


Figure 3. Frontal Systems Behavior Scale – T-scores before and after illness onset for Patient 2.

affected than Patients 1 and 2. Indeed, Patient 3's cognitive profile was also indicative of impairment in each of these additional areas: information processing speed, visual attention/scanning, working memory, semantic verbal fluency, visual perceptual abilities (see Figure 5), and verbal learning and memory. With regard to the latter, retrieval deficits were apparent given preserved performance on recognition tasks. This pattern of memory impairment is typical in conditions with frontal system and/or cerebellar involvement (see below).

Although some scores remained stable or demonstrated improvement (possibly because of practice), results of Patient 1's follow-up testing revealed declines in visual learning, delayed recall (verbal and visual), and complex working memory. Executive dysfunction remained evident both cognitively and behaviorally. Specifically, he was more apathetic, less self-aware, and the difficulties were having a greater negative impact on his family life, as per his wife's verbal report.

fMRI Results

Controls. Both functional tasks were performed in 10 healthy controls of similar age to Patients 1 and 2 (mean age 45.1 [9.42]) to establish standard patterns of blood flow during response inhi-

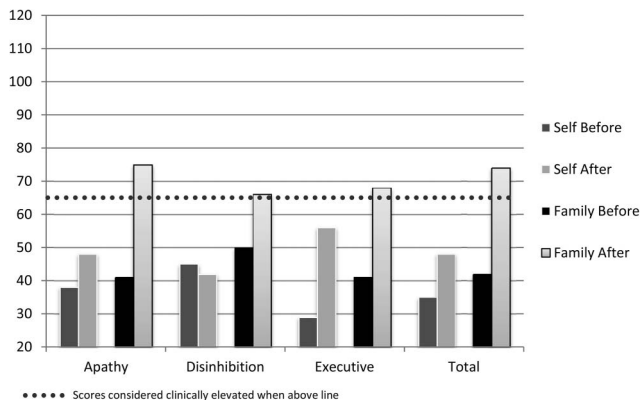


Figure 4. Frontal Systems Behavior Scale – T-scores before and after illness onset for Patient 3.

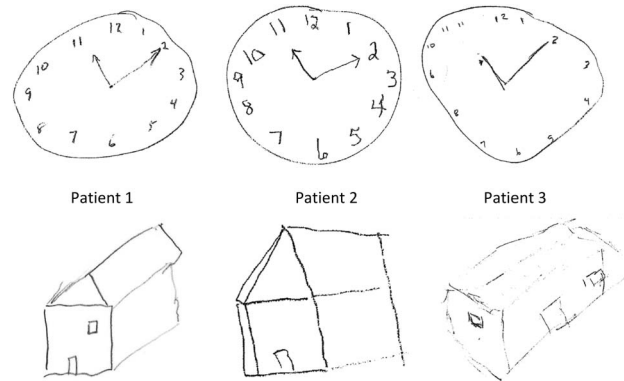


Figure 5. Clock and house drawings.

bition processing and working memory. The results were consistent with previous literature and included increased activity during the Go/NoGo task in bilateral inferior frontal gyri, anterior cingulate, premotor cortex, bilateral supramarginal gyri, and cerebellum. Increased activity was observed in the working memory task in middle and inferior prefrontal cortex, frontal poles, medial and lateral posterior parietal cortices, cingulate gyrus, and premotor cortices.

Patients-Go/NoGo task. Very little neural activation was observed in Patient 2 during the response inhibition task. However, Patient 1 had significant activation in the left inferior frontal gyrus, typical of response inhibition. There was also significant activity in a periventricular distribution that is not commonly observed in this type of task. This activity was lateralized to the left and quite substantial within the white matter, extending into the cingulate gyrus (see Figure 6).

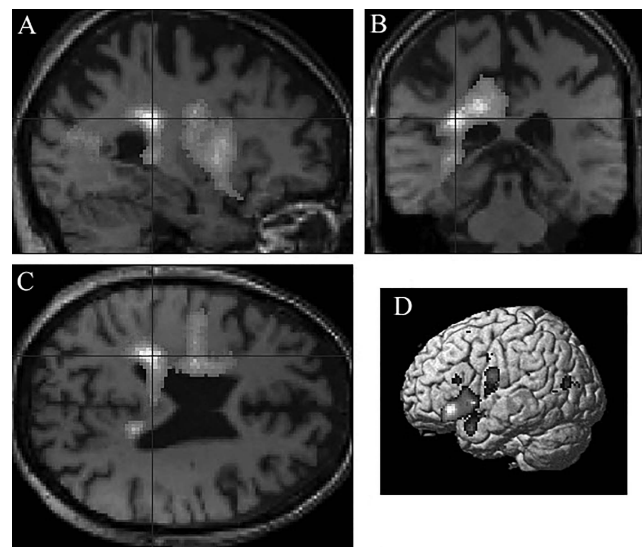


Figure 6. Response inhibition processing in Patient 1. (A) a sagittal view of the left periventricular white matter activation during the Go/No Go task, (B) a coronal view in neurological representation (left is left), and (C) an axial slice of the same activity. (D) represents a rendered image of the left inferior frontal gyrus activation during the same task.

Patients–N-back task. A similar pattern of periventricular white matter activity extending into the cingulate gyrus was observed in Patient 2 during the working memory task. Patient 2 activated the superior parietal lobule, as did Patient 1, but also activated the left middle frontal gyrus, as expected during working memory processing. Patient 1 had no prefrontal activation during the working memory task but rather relied solely on the more posterior parietal, temporal, and cerebellar regions (see Figure 7).

Both participants showed reduced FA and increased MD, suggesting impaired white matter integrity. The histograms and images show more white matter alterations in Patient 1 than Patient 2 (Figures 8 and 9).

Discussion

Some aspects of cognition in all three individuals remain within normal limits. Indeed, Patient 1 demonstrated relatively preserved functioning in intellectual functions, basic attention and mental control, scanning/sequencing, visual perceptual abilities, most aspects of learning and memory, and a number of areas of executive functioning as noted above. The presentation of Patient 2 was similar with the additionally preserved verbal fluency. Patient 3 was more compromised and thus performance fell within normal limits only on basic attention and crystallized

intellectual functions. When examining the areas of cognitive compromise, it is clear that all three individuals demonstrate some executive difficulties that are consistent with frontal-system dysfunction. While not all aspects of executive functioning were affected, Patient 1 and 2 demonstrated a vulnerability to perseveration, reduced mental flexibility, and diminished working memory under conditions of interference. Patient 3 had more globally impaired functioning. When these findings are coupled with the predominance of the dysexecutive behavioral presentation (i.e., all three subscales of the FrSBe reaching clinical significance for informant ratings), findings suggest that executive dysfunction is a prominent area of cognitive dysfunction in all three individuals. Although the presumed declines from estimated average premorbid ability (based on educational and occupational functioning as well as current performance on IQ measures) were subtle on some tasks, the pattern of dysfunction (particularly given the behavioral presentation) remains consistent with a dysexecutive profile. Sensory-motor and language functioning were also negatively impacted. ADCA-DN patients are known to have cognitive and behavioral deficits; however, these deficits have not previously been well-defined with extensive neuropsychological testing. Judgment and executive function problems and behavior dysfunction leading to a diagnosis of probable FTD has been described (Klein et al., 2013) for patients with the related disorder HSAN1E, although again, details of specific test findings are lacking in the literature. Whereas all three of the current individuals demonstrated executive dysfunction, word-finding difficulties, and reduced fine motor abilities, Patient 3's cognition was also more globally affected with deficits also noted in complex attention, speed of processing, visual perceptual abilities, and learning/memory. Patients 1 and 2 are likely earlier in the disease course and also demonstrate milder cognitive impairment, exhibiting some similarities with individuals with preclinical frontotemporal lobar degeneration (FTLD). One could also characterize the clinical presentation as being similar to that observed in early *cerebellar cognitive affective syndrome* (CCAS; also called *Schmahmann's syndrome*; Manto & Marien, 2015; Schmahmann & Pandya, 1997). In addition to the cerebellar contribution to the ataxic gait, this structure, along with its intricate feedback loops with both basal ganglia and frontal lobe structures (revealed by functional connectivity studies; Krienen & Buckner, 2009), has a substantial impact on cognitive functioning as well. Thus, the “dysmetria of movement is then matched by an unpredictability and illogic to social and societal interaction” (Schmahmann & Pandya, 1997, p. 55). Case studies and anatomical investigations of this condition have documented that behavioral and neuropsychiatric changes were most prominent in those with involvement of the posterior lobe of the cerebellum and vermis (Schmahmann & Sherman, 1998; Stoodley & Schmahmann, 2010). The “frontal” nature of the cognitive and behavioral dysfunction documented currently is then likely because of the basal ganglia and cerebellar loops with the cerebral cortex that serve cognitive functions (Middleton & Strick, 1994). This is supported by the cerebellar activation noted in Patient 1 during the working memory fMRI task.

Given the extent and severity of Patient 3's cognitive dysfunction, as well as his reported functional impairment, he does meet clinical criteria for a diagnosis of dementia; currently

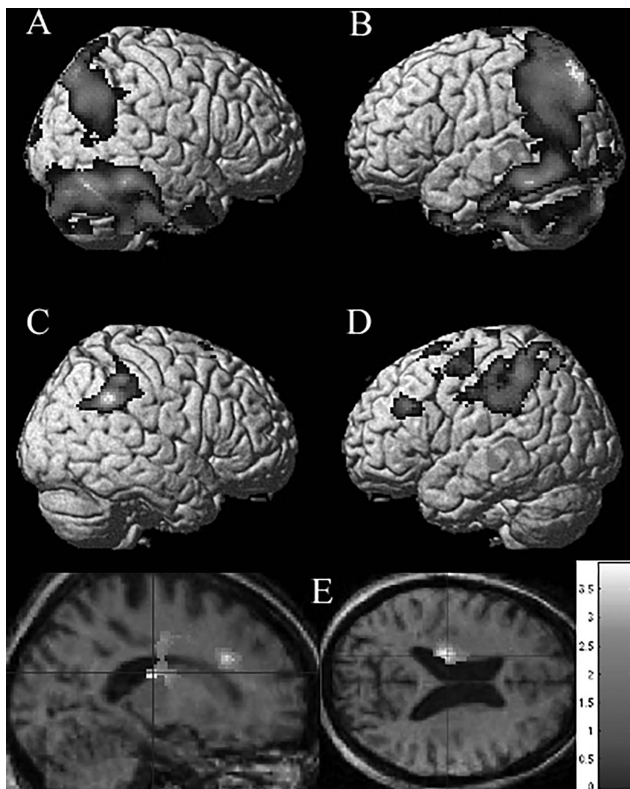


Figure 7. Working memory processing in both participants. A and B represent the rendered images of posterior brain activity observed in Patient 1 during the two-back task. C and D represent the same for Patient 2. Similarly, E represents the left periventricular white matter activity in Patient 2 during the same task.

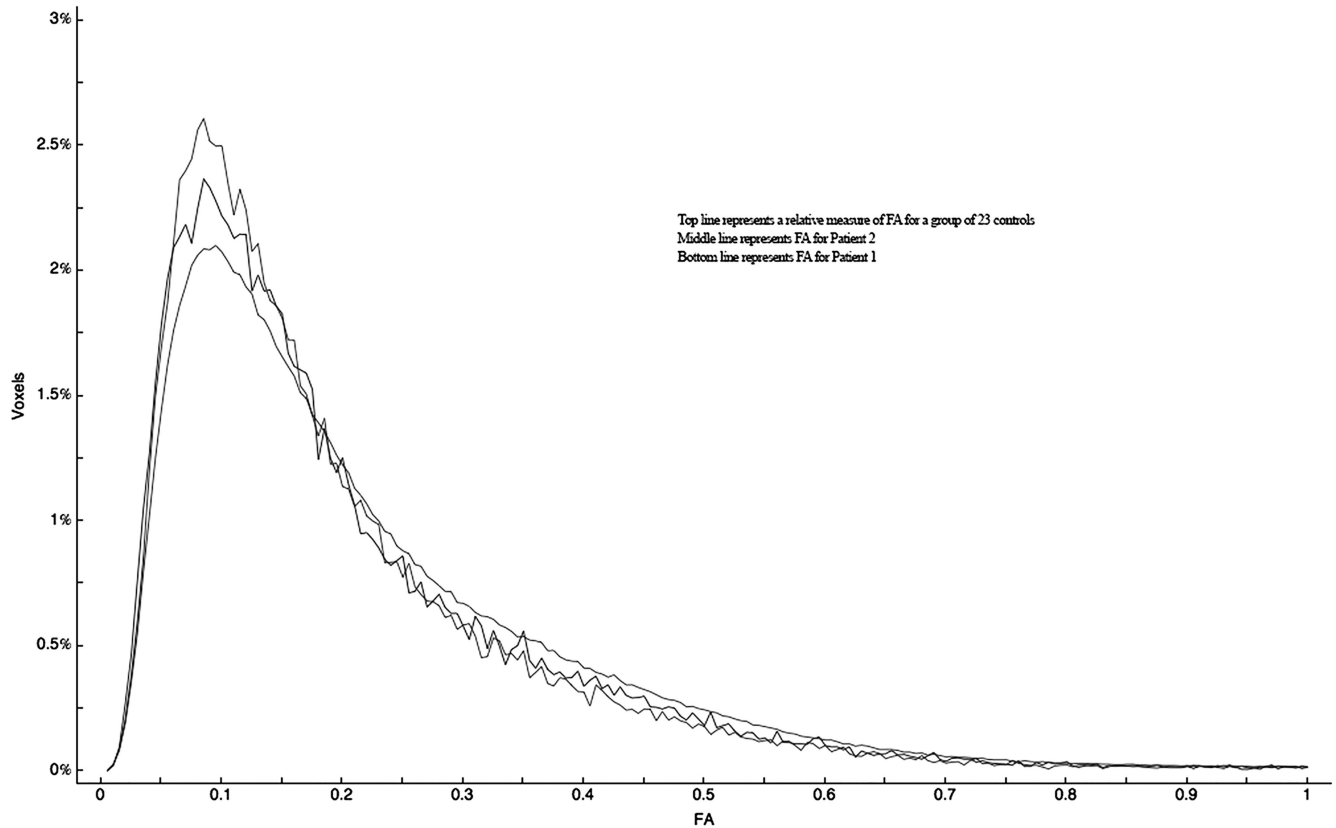


Figure 8. DTI histogram of results for fractional anisotropy (FA). The top line represents a relative measure of FA for a group of 23 controls, the bottom line is Patient 1 and the middle line is Patient 2.

termed *major neurocognitive disorder* under Diagnostic and Statistical Manual for Mental Disorders-Fifth Edition (*DSM-5*) criteria, given that he has demonstrated a significant decline from a previous level of performance in more than one cognitive domain and that he now requires assistance with daily activities (American Psychiatric Association, 2013). If we can consider Patient 3 as being further along in the disease course (given his older age and his more severe physical and cognitive symptomatology), we can speculate that the cognitive impairment in *DNMT1* mutation likely impacts some aspects of executive functioning initially (as well as sensory-motor and language function), but eventually progresses to a more global cognitive impairment. The cognitive decline appears to proceed in parallel with the physical decline.

In the past, ADCA-DN and HSN1E were considered to be distinct conditions with *DNMT1* mutations on exons 21 and 20, respectively. In ADCA-DN, hearing loss has been present, but it has typically been more minor and occurred later in the disease course compared with HSN1E. Early symptoms of ADCA-DN have included narcolepsy, cataplexy, and later ataxia. Narcolepsy was not typically observed in HSN1. However, recently Moghadam and colleagues (2014) performed phenotypic characterization in two probands from two ADCA-DN Italian kindreds and three probands from two Italian HSN1E kindreds to allow for a more direct comparison between the two conditions (Moghadam, Pizza, Morgia et al.,

2014). They noted that narcolepsy (with or without cataplexy) was present in both phenotypes. They described early features of ADCA-DN, in young and otherwise asymptomatic individuals, as foreshadowing the onset of the illness including: excessive daytime sleepiness, sleep onset REM periods (Moghadam, Pizza, Tonon et al., 2014) and cataplexy. Excessive sleepiness (Klein et al., 2013) and narcolepsy (Baets et al., 2015) have been described in HSN1E. In addition to narcoleptic features, ADCA-DN and HSN1E case descriptions share several common signs: subclinical optic neuropathy, deafness, cerebellar ataxia, and large and small fiber neuropathy (Moghadam, Pizza, Morgia et al., 2014). Supporting evidence for overlap of phenotypic presentation was found in a sporadic Japanese case of *DNMT1* mutation on exon 21 whose phenotype resembled that of HSN1E (Yuan et al., 2013). Moghadam and colleagues (2014) conclude that ADCA-DN and HSN1E represent two discrete clinical entities, but along the same disease spectrum. The cognitive presentation of the currently evaluated ADCA-DN kindred also documented an overlap in cognitive profile with those with *DNMT1* mutation causing HSN1E, providing further support to the notion that these conditions are part of the same clinical disease spectrum. Although we do not have the benefit of direct comparisons with the same cognitive test battery, current findings suggest a remarkable similarity given that some aspects of frontal-executive function are affected.

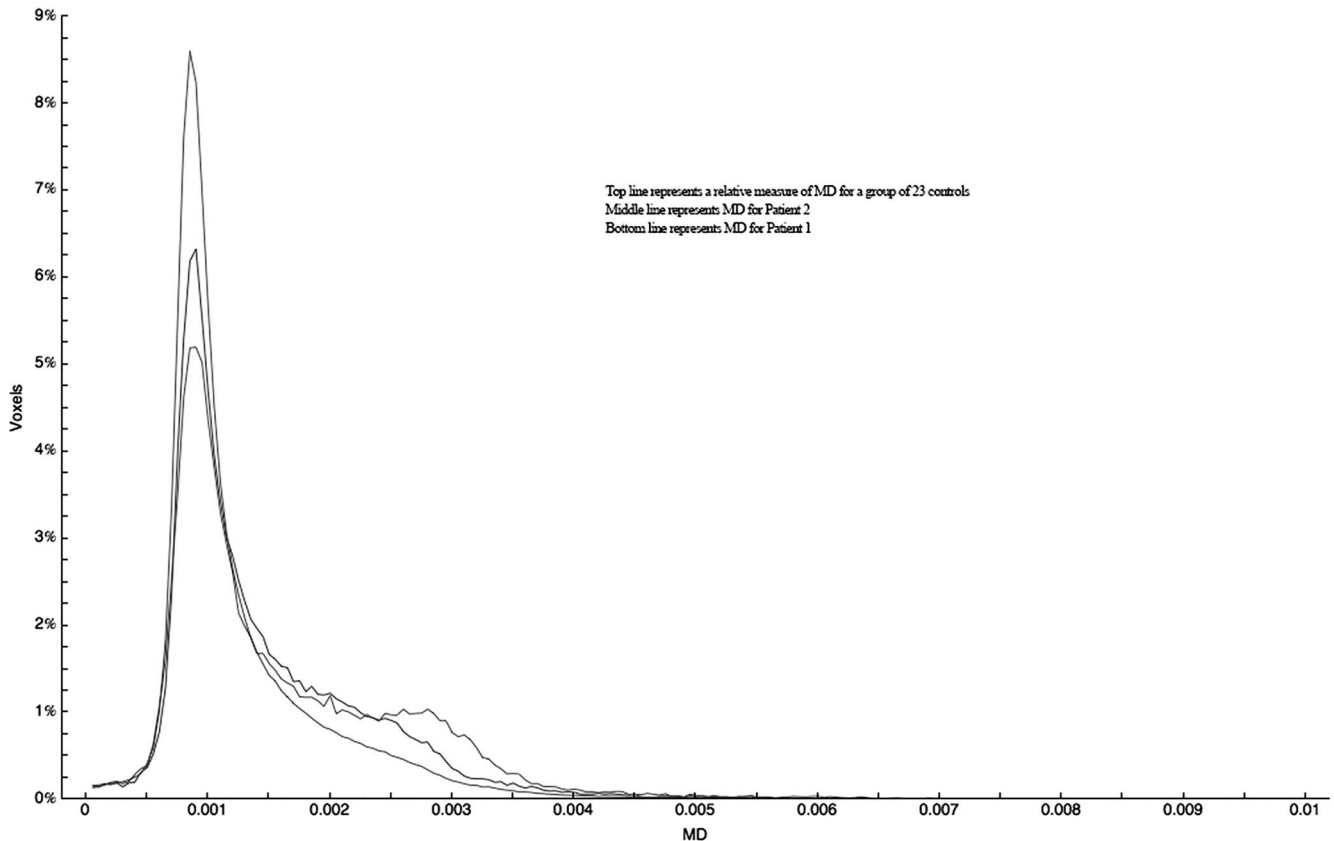


Figure 9. DTI histogram of results for mean diffusivity (MD). The top line represents a relative measure of MD for a group of 23 controls, the bottom line is Patient 1 and the middle line is Patient 2.

With regard to the fMRI findings, the working memory task required more cerebral activity in both participants, suggesting more effortful processing. This is corroborated by the neuropsychological testing that showed both participants as impaired on working memory but not for response inhibition. In addition, Patient 1 did not engage his prefrontal cortex during working memory, relying solely on excessive activity of his parietal and temporal lobes and his cerebellum to perform the task. The cerebellar involvement is consistent with previous topographical research demonstrating a role for the cerebellum in cognitive processing (Stoodley & Schmahmann, 2010). Patient 2 showed a more common pattern of activity for the working memory task. Both participants showed a considerable amount of periventricular white matter activity which was not anticipated for these tasks. The DTI results show that the integrity of the white matter is negatively impacted in both participants. This could explain the altered pattern of neural activity during the functional tasks. The white matter atrophy observed in Patient 1 compared with Patient 2 may also explain the observed differences.

In summary, although *DNMT1* mutations can cause two distinct phenotypes, they are very likely part of the same disease spectrum with more symptom overlap than previously thought. The cognitive phenotype of the ADCA-DN subtype had not been previously characterized in the literature. The results from the current kindred

suggest an initial prominence of executive dysfunction that progresses to a more global dementia. The cognitive decline appears to proceed in parallel with the neurological decline.

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